Question:

Calculate the quantity of heat required to (a) convert a liter of H_2O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

A. 28 Kcal, 6.3 Kcal

B. 40 Kcal, 5 Kcal

C. 30 Kcal, 8.58 Kcal

D. 25 Kcal, 8 Kcal

E. 29 Kcal, 7.77 Kcal

F. 34 Kcal, 6.9 Kcal

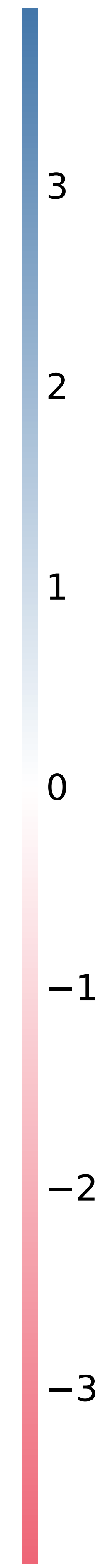
G. 30 Kcal, 6.45 Kcal

H. 32 Kcal, 7 Kcal

I. 35 Kcal, 7.5 Kcal

J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:



<think>

For part (a), we use the formula: q = m * c * ÎĶT

- Water mass (m) = 1000g (1L)

- Specific heat of water (c) = 1 cal/gÂ°C

- Temperature change (ÎĶT) = 60Â°C - 30Â°C = 30Â°C

- q = 1000 * 1 * 30 = 30,000 cal or 30 Kcal

For part (b), the mass of aluminum (m) = 1000g, specific heat of aluminum (c) = 0.215 cal/gÂ°C, and the temperature change (ÎĶT) = 60Â°C - 30Â°C = 30Â°C

- q = 1000 * 0.215 * 30 = 6450 cal or 6.45 Kcal

Therefore, the correct answer is C. 30 Kcal, 6.45 Kcal.

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<answer>C</answer>